

Verdant Coil – Scene & System Tree Diagram

This diagram outlines the high-level scene structure and system architecture for Verdant Coil, following Godot best practices and modular design.

Main.tscn

```
Main.tscn
├─ MainMenu.tscn
├─ ExploreMode.tscn
│   ├─ TileMap / TileContainer
│   ├─ Crawler.tscn
│   │   └─ UpgradeController.gd
│   ├─ TurnManager.gd (autoload)
│   └─ UI (CanvasLayer)
│       ├─ ResourceBar.tscn
│       ├─ UpgradeBar.tscn
│       └─ ActionButtons.tscn (Restart, Menu)
├─ BuilderMode.tscn
│   ├─ TileGridEditor
│   ├─ BiomassManager.gd
│   ├─ LevelValidator.tscn
│   └─ UI (CanvasLayer)
│       ├─ ResourceBar.tscn
│       ├─ TilePalettePanel.tscn
│       ├─ ActionButtons.tscn (Validate, Save, Publish, Undo)
│       └─ Tooltip.tscn
├─ LevelBrowser.tscn
│   ├─ MetadataViewer.tscn
│   └─ LevelImporter/Exporter.gd
├─ SettingsMenu.tscn (optional)
└─ ProfilePanel.tscn (optional)
```

Supporting Systems (Autoloads or Globals)

```
TurnManager.gd
GameStateManager.gd
PlayerProfile.gd
```

```
LevelDataManager.gd
ProjectileManager.gd
UpgradeController.gd
AnalyticsLogger.gd (post-MVP)
```

✨ Tile Variants (instantiated into grid at runtime)

```
BaseTile.gd
├─ FleshTile.tscn
├─ AcidPoolTile.tscn
├─ DigestWallTile.tscn
├─ StickyWebTile.tscn
├─ HeartrootTile.tscn
├─ PulseGateTile.tscn
├─ TriggerNodeTile.tscn
├─ GuardNoduleTile.tscn
├─ SpineShooterTile.tscn
└─ TendrillashTile.tscn
```

🍁 Debug / Development Tools

```
DevConsole.tscn
├─ UnlockAll.gd
├─ TestPathfinder.gd
└─ ManualResourceAdjuster.gd
```

Post-MVP Online Infrastructure

```
OnlineLevelSync.tscn
├─ BackendAPI.gd
├─ AccountManager.tscn
└─ OnlineLeaderboard.tscn
```